A career in Data Analysis

Welcome to this introductory course on Data Analytics, the first in a series of courses designed to prepare you for a career as a Junior Data Analyst.

A career in Data Analysis

"Businesses today recognize the untapped value in data and data analytics as a crucial factor for business competitiveness. To drive their data and analytics initiatives,companies are hiring and upskilling people. They're expanding their teams and creating centers of excellence to set up a multipronged data and analytics practice in their organizations."

-The Power Of Data To Transform Your Business, a Forrester report

https://www.ibm.com/downloads/cas/1NOA5OXZ

A career in Data Analysis

Combined to this is the significant supply and demand mismatch in skilled data analysts, making it a highly sought after and well-paid profession.

A career in Data Analysis

Mastering data analytics as a career path

Associate Data Analyst

Data Analyst

Senior Data Analyst

Lead Analyst

Principal Analyst

A career in Data Analysis

Branching into other data professions

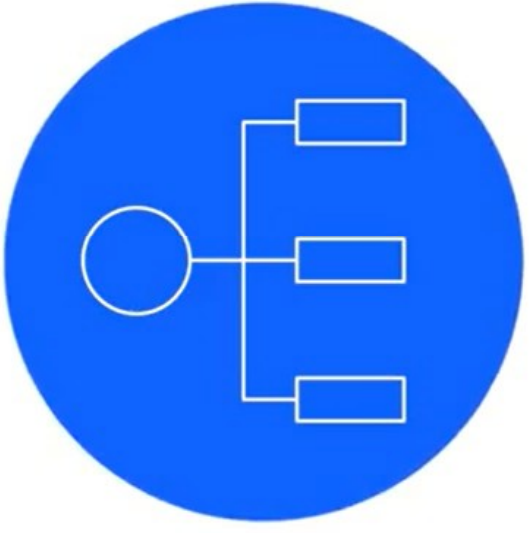
- Data Science
- Business Analytics
- Data Engineering
- Business Intelligence Analytics

A career in Data Analysis

- Fresh Graduate
- Working Professional considering a mid-career transition
- Data-driven Decision-maker
- Analytics-enabled job role

A career in Data Analysis

The course introduces you to the core concepts, processes, and tools you need to



- Gain entry into data analytics
- Strengthen your current role as a data-driven decision-maker

A career in Data Analysis

The course will equip you with an understanding of

- Data ecosystem
- Fundamentals of data analysis, such as data gathering, wrangling, mining, analysis, and data visualization
- A day in the life of a Data Analyst

A career in Data Analysis

Practicing data analysts share

- Their experience in gaining entry into this field
- Career options and learning paths you can consider
- What employers look for in a data analyst
- Their knowledge and best practices about the data analysis process

A career in Data Analysis

Congratulations on choosing to be on this journey

Good luck!

Modern Data Ecosystem

SKILLS NETWORK

IBM **Developer**

Modern Data Ecosystem

"The constant increase in data processing speeds and bandwidth, the nonstop invention of new tools for creating, sharing, and consuming data, and the steady addition of new data creators and consumers around the world, ensure that data growth continues unabated. Data begets more data in a constant virtuous cycle."

Forbes 2020 Report

Modern Data Ecosystem

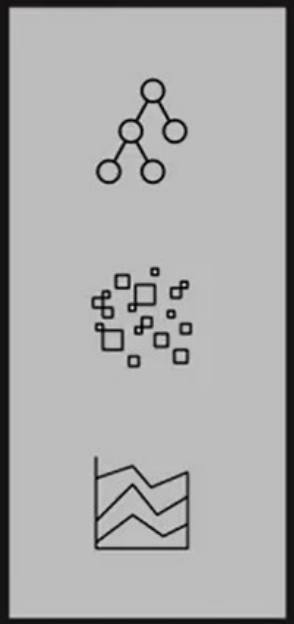
Interconnected

Independent

Continually evolving

Modern Data Ecosystem

Data Sources

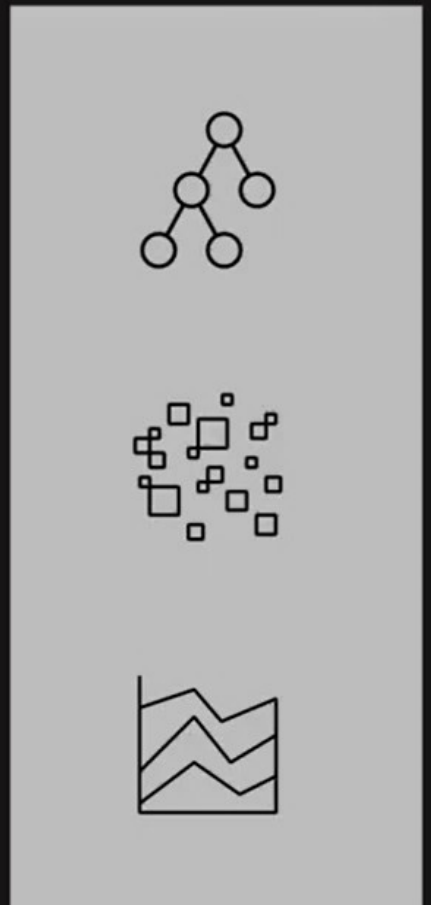


ENTERPRISE DATA ENVIRONMENT

- Data integrated from disparate sources
- Different types of analysis and skills to generate insights
- Active stakeholders to collaborate and act on insights generated
- Tools, applications, and infrastructure to store, process, and disseminate data as required

Users

Data Sources



Structured

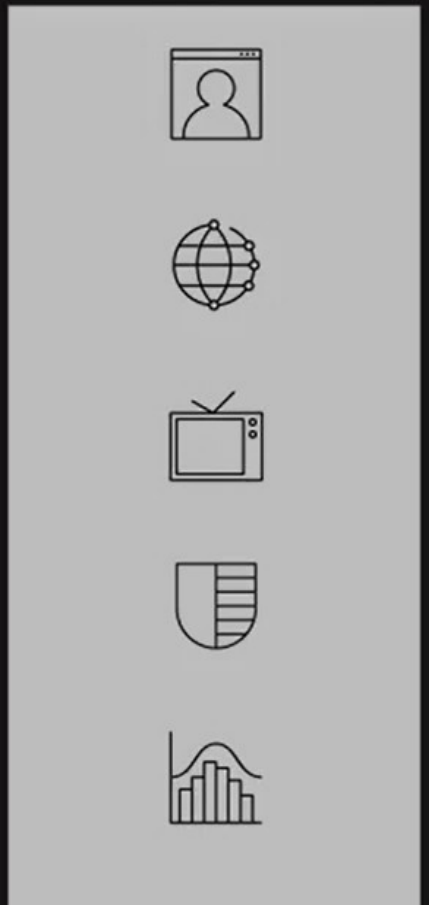
Unstructured

Data Sources

Structured

Unstructured

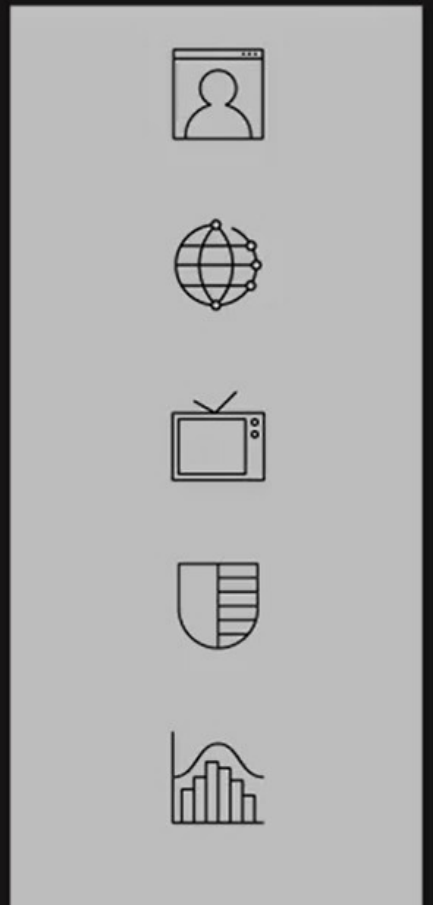
Data Sources



Structured

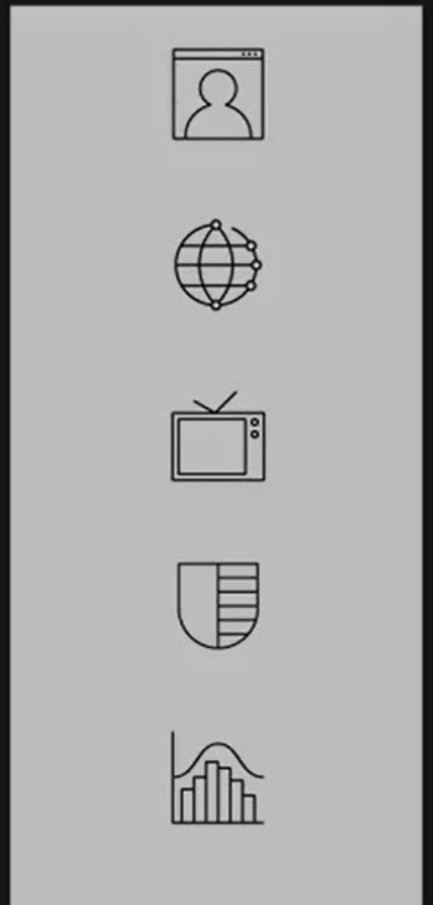
Unstructured

Data Sources



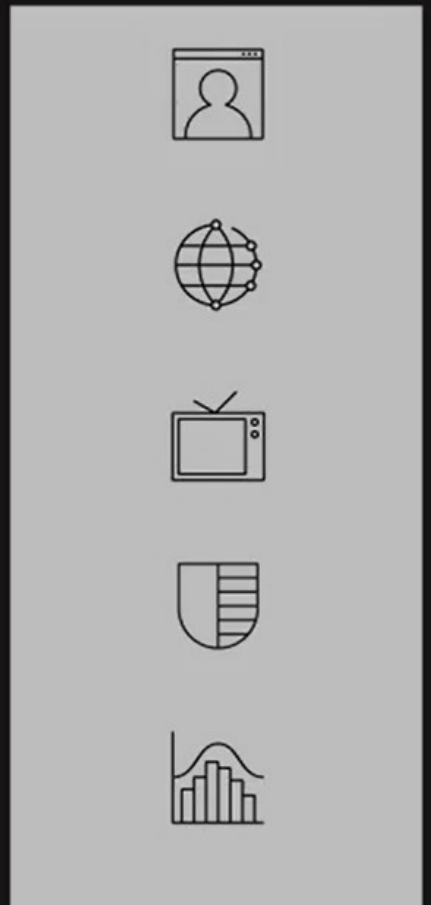
The first step is to pull a copy of the data from the original sources into a data repository

Data Sources



Acquiring the data you need, working with data formats, sources, and interfaces through which this data can be pulled.

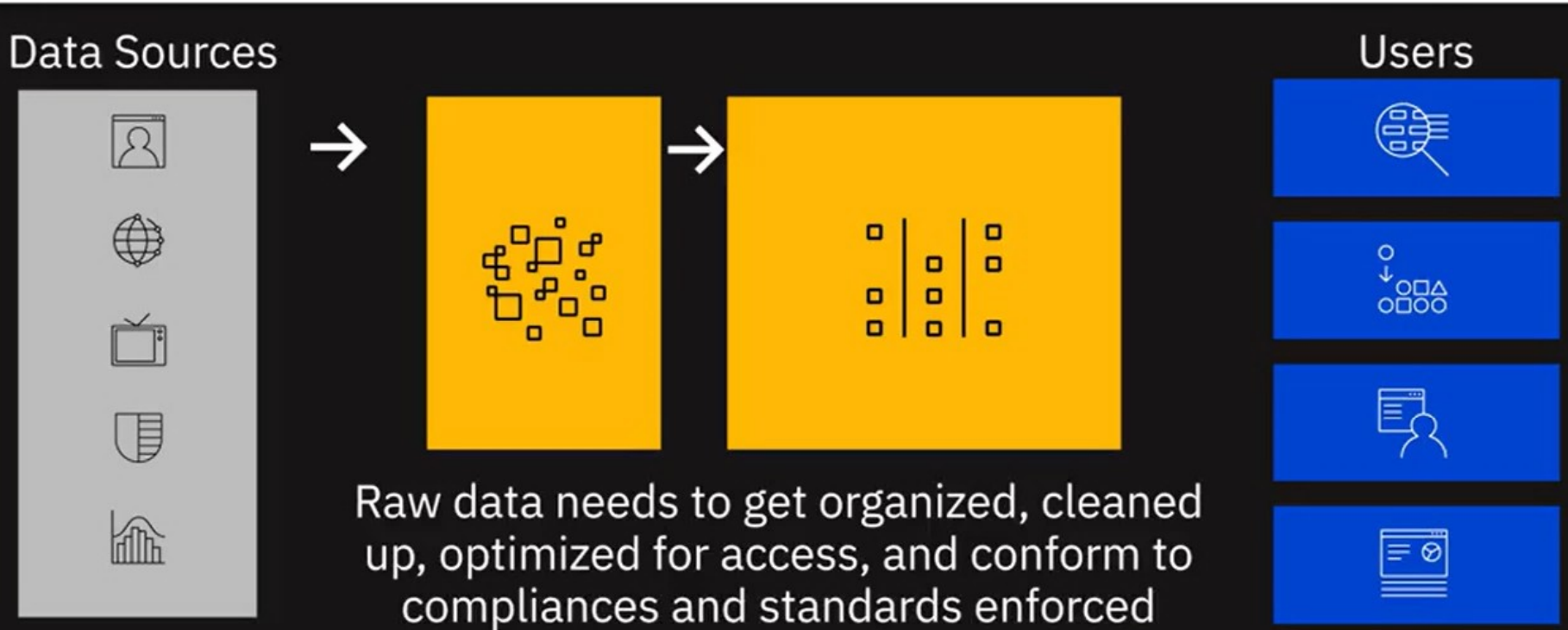
Data Sources

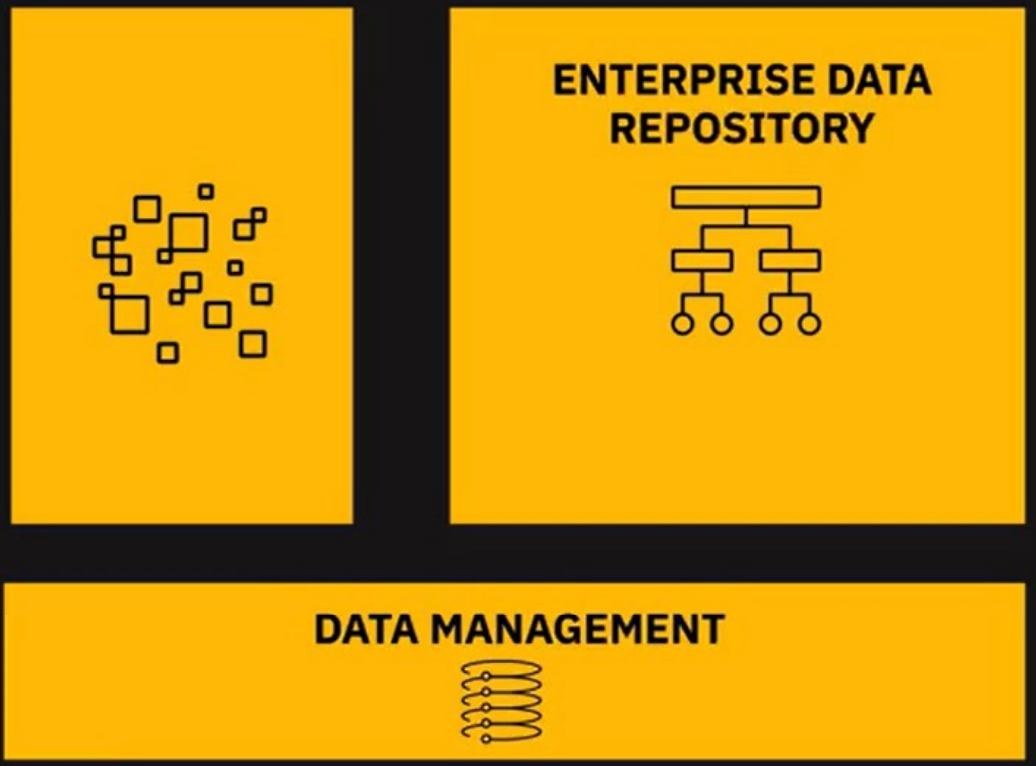


Challenges:

Reliability, security, and integrity of the data

Enterprise Data Environment

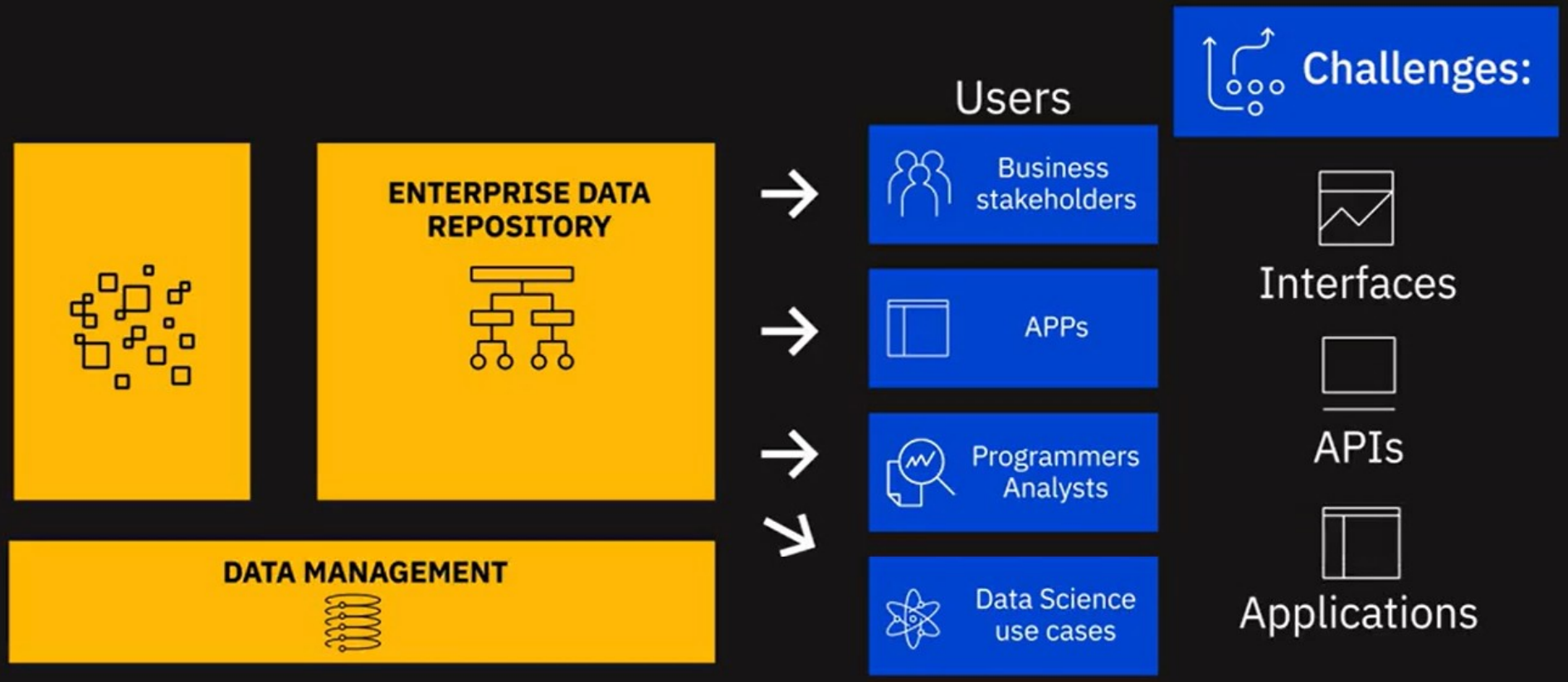




Challenges:

Data management

Repositories that provide high availability, flexibility, accessibility, and security


ENTERPRISE DATA REPOSITORY
DATA MANAGEMENT
Users
Business stakeholders
APPs
Programmers Analysts
Data Science use cases
Challenges:
Interfaces
APIs
Applications

Emerging technologies shaping the modern data ecosystem

Cloud Technologies

Machine Learning

Big Data

Emerging technologies shaping the modern data ecosystem

- Every enterprise today has access to limitless storage, high-performance computing, open source technologies, machine learning technologies, and the latest tools and libraries
- Data Scientists are creating predictive models by training machine learning algorithms on past data
- Big Data is paving the way for new tools and techniques and also new knowledge and insights

Key Players in the Data Ecosystem

SKILLS NETWORK

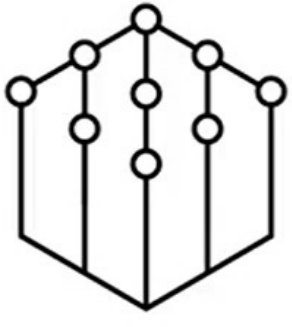
IBM **Developer**

To summarize

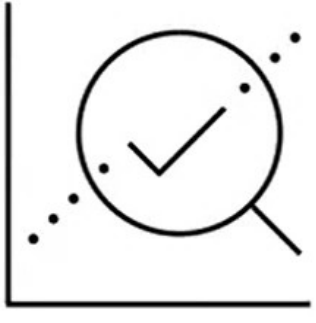
- Data Engineering convert raw data into usable data
- Data Analytics use this data to generate insights
- Data Scientists use Data Analytics and Data Engineering to predict the future using data from the past
- Business Analysts and Business Intelligence Analysts use these insights and predictions to drive decisions that benefit and grow their business

What is Data Analysis?

Gather, clean, analyze, and mine data

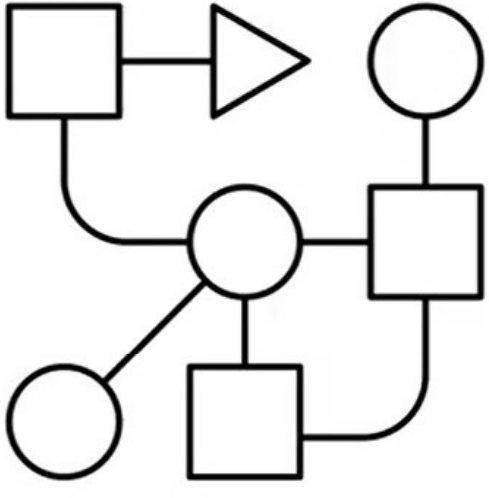


Interpret results



Report findings

Patterns
Correlations



Insights
Conclusions

Data Analysis helps businesses

- Understand past performance
- Take informed decisions
- Validate course of action—saving time and resources, ensuring success

Different types of Data Analysis

Descriptive Analytics

"What happened"

- Provides insights into past events

Diagnostic Analytics

"Why did it happen"

- Takes the insights from descriptive analytics to dig deeper to find the cause of the outcome

Predictive Analytics

"What will happen next"

- Leverages historical data and trends to predict future outcomes

Prescriptive Analytics

"What should be done about it"

- Analyzes past decisions and events to estimate the likelihood of different outcomes

The Data Analysis Process

Understanding the problem and desired result

Defining where you are and where you want to be

Setting a clear metric

Deciding what will be measured and how it will be measured

Gathering data

Identifying data you require, the sources from which you will access this data, and the best tools for the job

The Data Analysis Process

Cleaning data

Fixing quality issues in the data and standardizing data coming in from multiple sources

Analyzing and Mining data

Extracting, analyzing, and manipulating data from different perspectives to understand trends, identify correlations, and find patterns and variations

Interpreting results

Interpreting results, evaluating defendability of analysis and circumstances under which analysis may not hold true

The Data Analysis Process

Analyzing and Mining data

Extracting, analyzing, and manipulating data from different perspectives to understand trends, identify correlations, and find patterns and variations

Interpreting results

Interpreting results, evaluating defendability of analysis and circumstances under which analysis may not hold true

Presenting your findings

Communicating and presenting your findings in clear, impactful, and convincing ways